



Lipidomics of Copper Stress on *Arabidopsis Thaliana*

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How Does Lipid Composition Change from Nutrient Stress?

Copper is one of eight essential plant micronutrients, in which copper plays a critical role in plant development, such as photosynthesis, protein synthesis, and respiration. The exposure of excess copper can result in some defects in plant metabolism; as a result, plants have a complex network of metal trafficking pathways to appropriately regulate copper homeostasis (Saglam et al, 2016).

These defense mechanisms can include alterations in the content of lipids. Lipids are essential components of plants because they function as structural components for membranes and are important intracellular signals (Reszcynska and Hanaka, 2020). Because cell membranes are major targets of environmental stresses and lipids are important membrane components, the quantitative and qualitative changes in the lipid composition can alter membrane fluidity and its function.

Therefore, it is important to present the selected lipid tissues with the aim of explaining the differences in the lipid composition, determining the mechanisms during the acquisition and distributions of copper, and discover new useful products that will help approach problems in maintaining plant growth for agriculture, health, and the environment.

BENEFITS OF HYDROPONICS



No Soil



Fast Growth



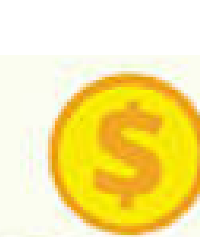
Less diseases



Less pesticide use



Higher Yields



Affordable

Maintaining plants for a period of time can be difficult, especially since there are many factors that contribute to its growth. Hydroponic culture allows us to have complete control on the specific amounts and type of compounds in the liquid growth medium. Moreover, this method resembles the uptake from soil without including the "flaws of potential chelation by soil components." (Pratelli et al, 2016)

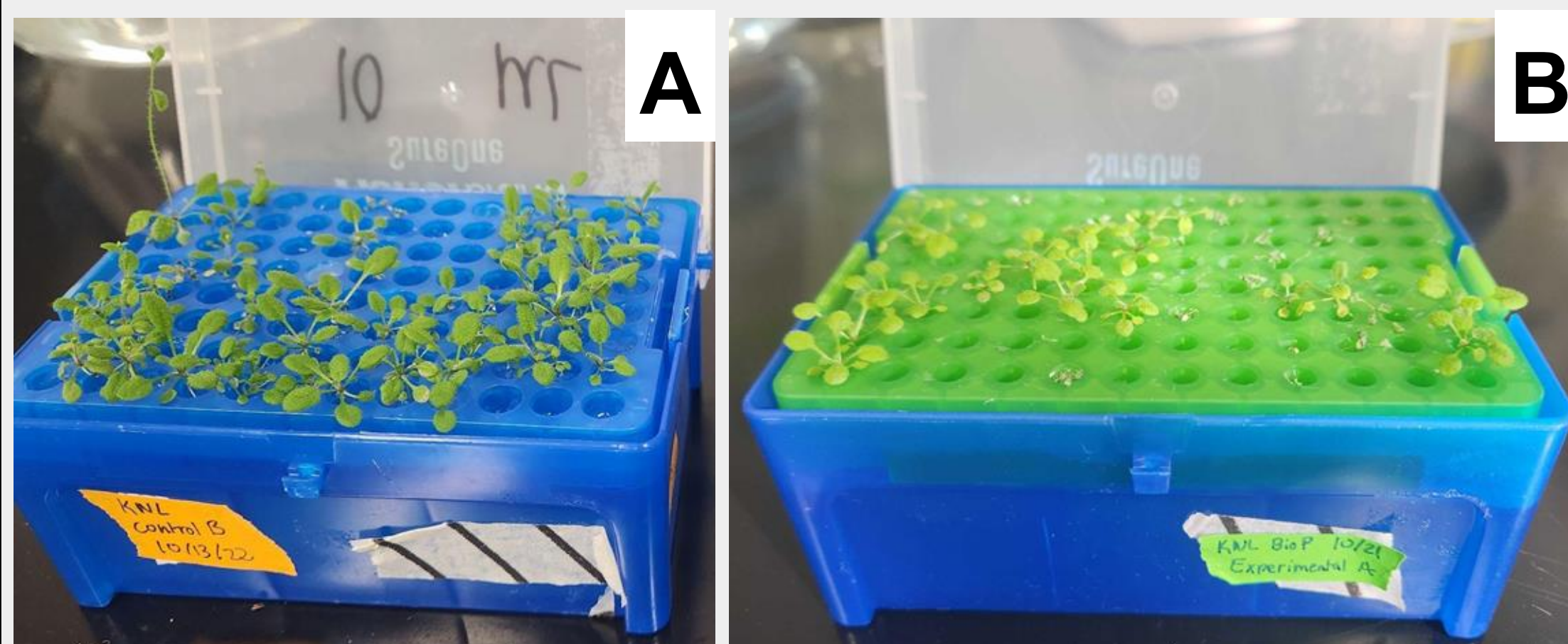


Figure 2. Hydroponics of *Arabidopsis thaliana* Between Control and Experimental Groups Hydroponic systems were utilized to grow *Arabidopsis thaliana* seeds. (a) The hydroponic on the left represents the control group on day 21. (b) The hydroponic on the right represents the experimental group on day 17.

Exploring the 'Omics of Copper Stress Responses

The *Arabidopsis thaliana* plants in the control group had roots that were approximately 7 cm long, darker pigmentations, and more plant biomass (Figures 2 and 3). On the other hand, the plants that were grown in excess copper had shorter roots (approximately 3 cm long), lighter pigmentations, and less plant biomass (Figures 2 and 3). These changes in morphology indicates that copper toxicity can have an effect in plant growth and development. In fact, the gradual accumulation of copper in plant tissues can be accompanied by specific changes in lipid composition, such as lipid peroxidation of about 80% of lipids in plants like monogalactosyl diacylglycerol (MGDG), digalactosyl diacylglycerol (DGDG), sulfoquinovosyl diacylglycerol (SQDG), and phosphatidylglycerol (PG) (Figure 4) (Rozentsvet et al, 2012). MGDG and DGDG are galactolipids, which are primarily found in the thylakoid membrane and play a major role in light-dependent photosynthesis. SQDG and PG are anionic lipids, which are cell signaling molecules that regulate fundamental cellular process like cell division, cell signaling, and cell growth (Kobayashi, 2016). The degradation of these lipids can affect the photosynthesis of plants, which ultimately stunts cell growth and development.

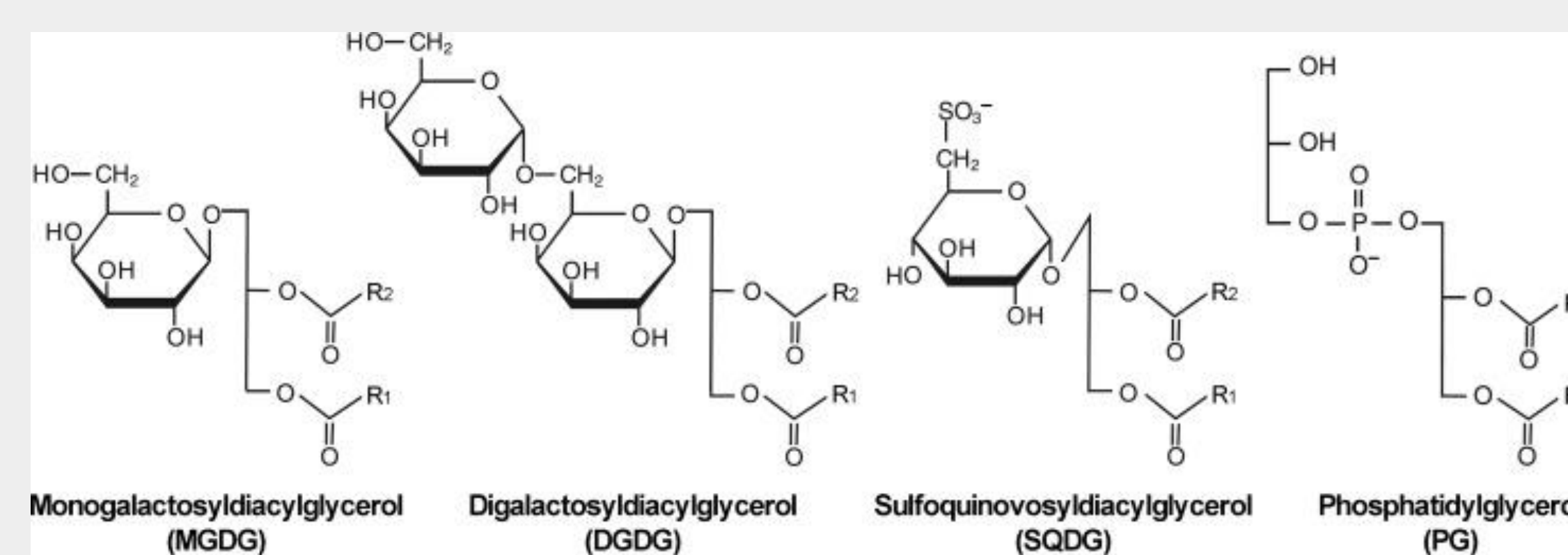


Figure 4. Structure of Major Lipids in *Arabidopsis Thaliana*



Figure 3. Root Length Between Control and Experimental Groups These plants were harvested on November 9, 2022, and their root lengths were measured on day 12. (a) The plants in Control C had an average root length of 7 cm. (b) The plants in Experimental C had an average root length of 3 cm.

Model Organism: *Arabidopsis Thaliana*: Unravelling the interaction between an organism's stress response and the chemical and functional outcomes is very challenging. As a result, utilizing a model organisms, such as *Arabidopsis thaliana*, can simplify these challenges because of the ease of growth, well-studied genomes, and short generation times.



A. Hydroponic (Day 0-14) B. Hydroponic (Day 15/+) C. Plant tissue



D. Lipid Extraction

E. Liquid Chromatography - Mass Spectrometry

Figure 1. Schematic of Identifying Lipid Composition in Response to Copper Stress

(a) *Arabidopsis* seeds are sterilized and cultured in two separate groups on hydroponics. The experimental group contains excess copper (10x copper, which is 1.3×10^{-5} M Cu), while the control group contains normal nutrient levels (1x copper). (b) After two weeks, the tip racks are transferred to tip boxes. (c) The plant tissues are harvested and freeze dried in liquid nitrogen. (d) The lipids are extracted through phase separation from the MTBE/MeOH liquid-liquid extraction method. (e) The collection of lipids from the upper layer (left) is brought to the Brodbelt lab to profile lipids through LC-MS (right).

Future Directions: Since the results indicated that excess copper may have a detrimental effects on the root structure, pigmentation, and overall growth of *Arabidopsis thaliana* plants, this suggests the idea that the plant's mechanism of growth, such as photosynthesis, involves changes in the lipid composition of chlorophyll. As a result, one of the future endeavors include successfully getting data from LC-MS that can profile the lipids in the control and experimental *Arabidopsis thaliana* plants to further support the results from this experiment. In addition, since exposure to 10x of the concentration of copper clearly showed changes in plant development, future experiments may include analyzing changes in lipid composition based on how much excess copper is exposed.

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References:

Chen, Shili, et al. "Simultaneous Extraction of Metabolome and Lipidome with Methyl Tert-Butyl Ether from a Single Small Tissue Sample for Ultra-High Performance Liquid Chromatography/Mass Spectrometry." *Journal of Chromatography A*, vol. 1298, July 2013, pp. 9–16. ScienceDirect, <https://doi.org/10.1016/j.chroma.2013.05.019>.
Gigon, Agnès, et al. "Effect of Drought Stress on Lipid Metabolism in the Leaves of *Arabidopsis Thaliana* (Ecotype Columbia)." *Annals of Botany*, vol. 94, no. 3, Sept. 2004, pp. 345–51. PubMed, <https://doi.org/10.1093/aob/mch150>.
Kobayashi, Koichi. "Role of Membrane Glycolipids in Photosynthesis, Thylakoid Biogenesis and Chloroplast Development." *Journal of Plant Research*, vol. 129, no. 4, 2016, pp. 565–80. PubMed Central, <https://doi.org/10.1007/s12025-016-0827-y>.
Matyash, Vitali, et al. "Lipid Extraction by Methyl-Tert-Butyl Ether for High-Throughput Lipidomics." *Journal of Lipid Research*, vol. 49, no. 5, May 2008, pp. 1137–46. ScienceDirect, <https://doi.org/10.1194/jlr.D700041-JLR200>.
Mustapha, Sherif Ishola, and Yusuf Makari Isa. "Utilization of Quaternary Solvent Mixtures for Extraction of Lipids from *Scenedesmus Obliquus* Microalgae." *Cogent Engineering*, edited by Harvey Arellano-Garcia, vol. 7, no. 1, Jan. 2020, p. 1788877. Taylor and Francis+NEJM, <https://doi.org/10.1080/23311916.2020.1788877>.
Reszczyńska, Emilia, and Agnieszka Hanaka. "Lipids Composition in Plant Membranes." *Cell Biochemistry and Biophysics*, vol. 78, no. 4, 2020, pp. 401–14. PubMed Central, <https://doi.org/10.1007/s12013-020-00947-w>.
Rozentsvet, Olga A., et al. "The Effect of Copper Ions on the Lipid Composition of Subcellular Membranes in *Hydrilla Verticillata*." *Chemosphere*, vol. 89, no. 1, Sept. 2012, pp. 108–13. ScienceDirect, <https://doi.org/10.1016/j.chemosphere.2012.04.034>.
Saglam, Aykut, et al. "Chapter 2 - Copper Stress and Responses in Plants." *Plant Metal Interaction*, edited by Parvaiz Ahmad, Elsevier, 2016, pp. 21–40. ScienceDirect, <https://doi.org/10.1016/B978-0-12-803158-2.00002-3>.